

ZHUFAN GU

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EDUCATION

- **Trine University, Phoenix, AZ | *Master of Science in Information Studies*** **May 2025 – May 2027**
• Courses: Object-oriented Programming, Database, Systems Engineering Analysis
- **Santa Clara University, Santa Clara, CA | *Master in Computer Science and Engineering*** **Sep 2022 – Dec 2025**
• Courses: Data Structures & Algorithms, Operating Systems, Computer Networks, Distributed Systems
- **University of St. Andrews, St. Andrews, UK | *Master in Human Computer Interaction*** **Sep 2019 – Dec 2020**
• Courses: User-Centered Interaction Design, Information Visualization

EXPERIENCE

- **CodeDay Labs | Software Engineer Intern** **Jun 2024 – Aug 2024**
• Developed a test framework to validate logical correctness of a **Rust** algebraic simplification engine used for hardware verification (utilizing E-graphs and the Z3 SMT solver).
• Bridged the gap between high-performance Rust and engineering workflows using **PyO3** and **Maturin**, making the verification tool accessible for large-scale computations in Python.

PROJECTS

- **Pantry AI | Full-Stack Web Development** **Sep 2025 – Oct 2025**
• Architected a multimodal AI application, building frontend with **Next.js/TypeScript** and backend with **Python/FastAPI**.
• Integrated multiple **AI models** for ingredient image recognition, recipe generation, and an interactive chat workflow.
• Deployed serverless backend on **AWS Lambda** using **Docker** for cross-platform packaging, frontend on **Amplify**.
- **AppliTrack | Full-Stack Development** **Sep 2023 – Nov 2023**
• Engineered a full-stack job tracking application using the **MERN** stack (**MongoDB, Express.js, React, Node.js**).
• Implemented a backend with **JWT authentication**, role-based access control, and data validation middleware.
• Developed a frontend utilizing **React Router** and **TanStack Query**.

OPEN SOURCE EXPERIENCE

- **React-PDF** (9.5k stars)
• Elevated accessibility from 96 to 100 by adding aria-labels to PDF page annotation links, addressing issues identified by Google Lighthouse.
- **Open Energy Dashboard**
• Built a BTU conversion engine using **JavaScript**, enabling dual-unit energy reporting for federal compliance standards.
• Authored comprehensive unit and integration tests with **Mocha.js** to validate all conversion logic.
• Configured **Docker** containers to standardize the multi-developer environment, reducing setup issues by 40% and ensuring consistent builds.

SKILLS

Languages: Python, JavaScript/TypeScript, HTML/CSS, Java, Rust, C++
Frameworks & Libraries: React/Next.js, FastAPI, Node.js, Express.js
Databases: MongoDB, PostgreSQL, Oracle, SQL
Developer Tools: Docker, Git, Vercel, Render, Cursor, GitHub Copilot, Postman
UI/UX Design Tools: Axure, Figma, Balsamiq, Blender

HONORS AND REWARDS

- Zhejiang Government Scholarship (Top 3%)
- Excellent Graduates of Zhejiang Province